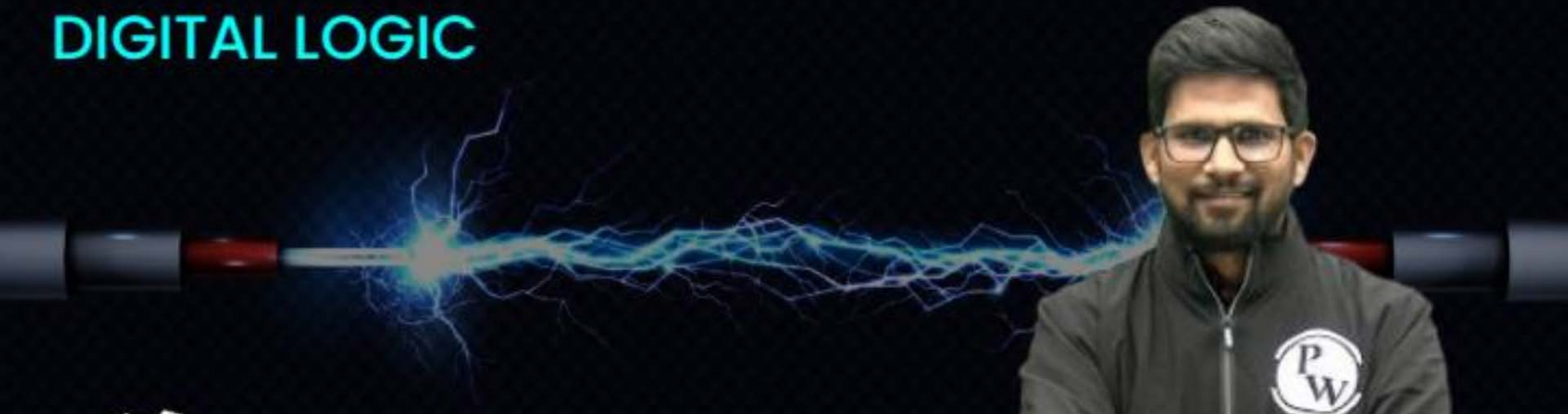
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 06

Sequential Circuit



By- Chandan Gupta Sir

Recap of Previous Lecture



- Asynch. Counter
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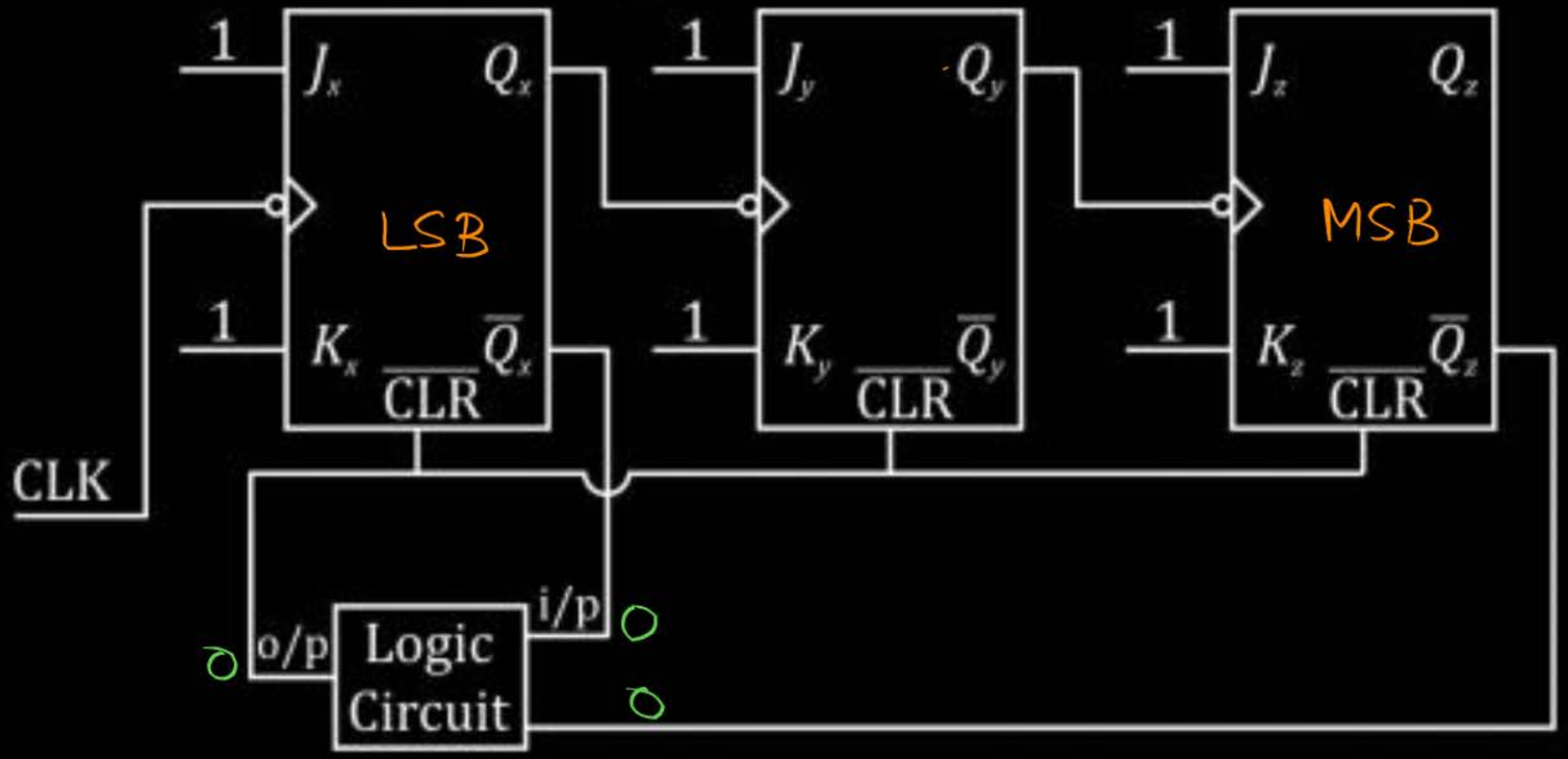


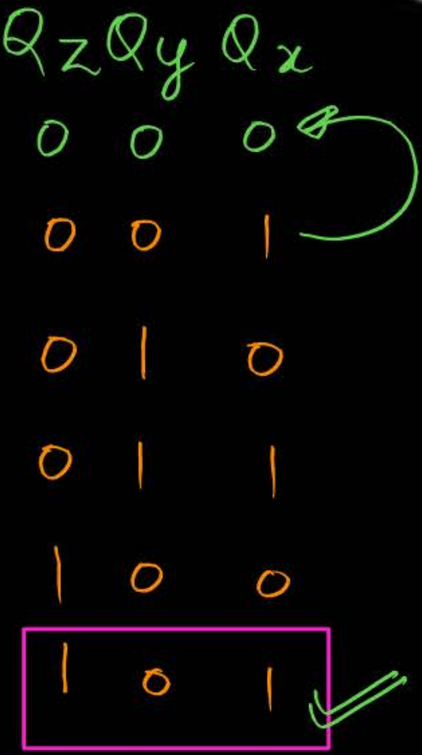
Asynch. Counter

Question

up counter

A sequential circuit is designed as shown :





For above circuit to work as MOD-5 counter, the logic circuit will be

(a) 2-input OR GATE

(b) 2-input AND GATE

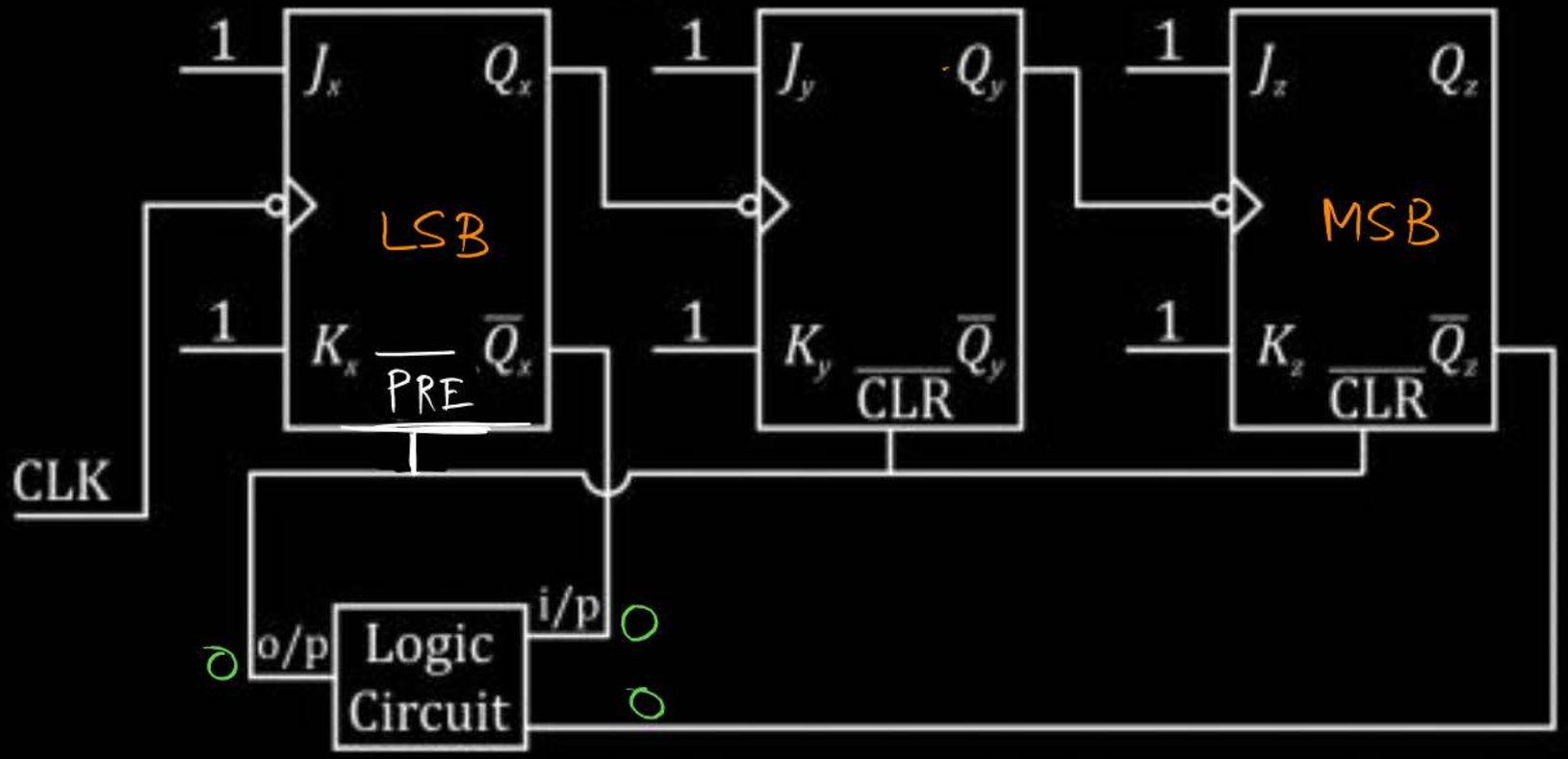
(c) 2-input NAND GATE

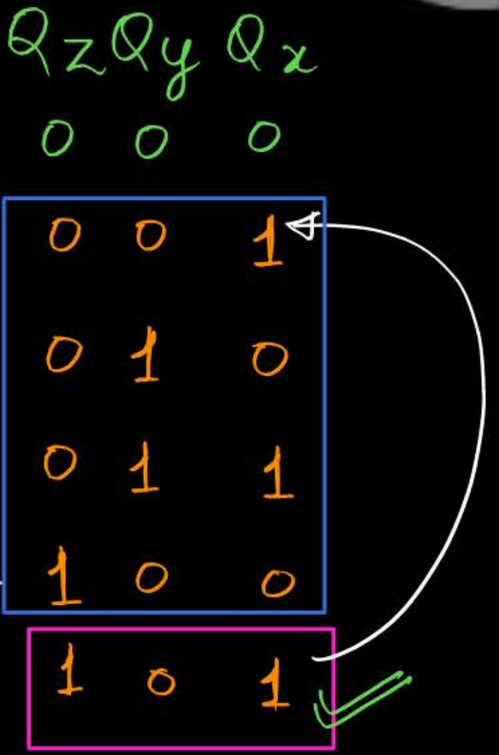
(d) 2-input NOR GATE

Question

up counter

A sequential circuit is designed as shown :





For above circuit to work as MOD-4 counter, the logic circuit will be

(a) 2-input OR GATE

(b) 2-input AND GATE

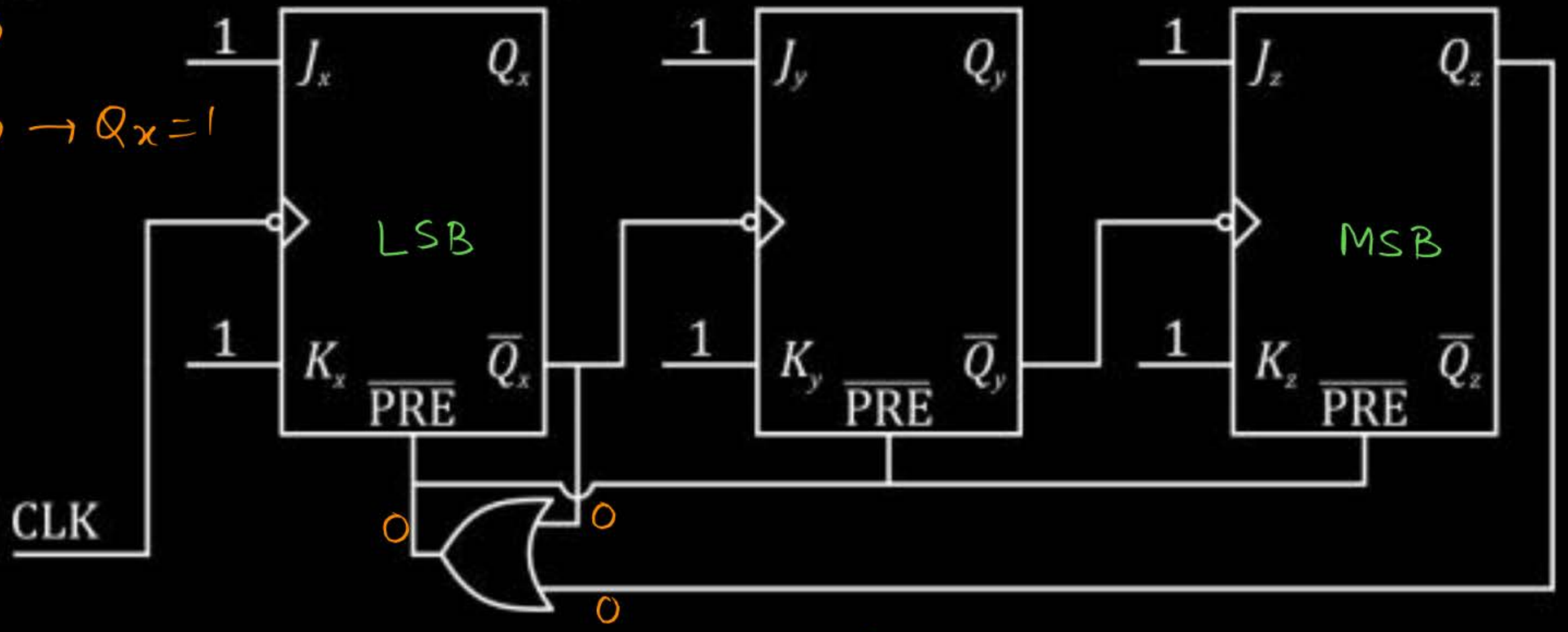
(c) 2-input NAND GATE

(d) 2-input NOR GATE

Question

down Counter

A sequential circuit is as given below :



$Q_z = 0$

$\overline{Q}_x = 0 \rightarrow Q_x = 1$

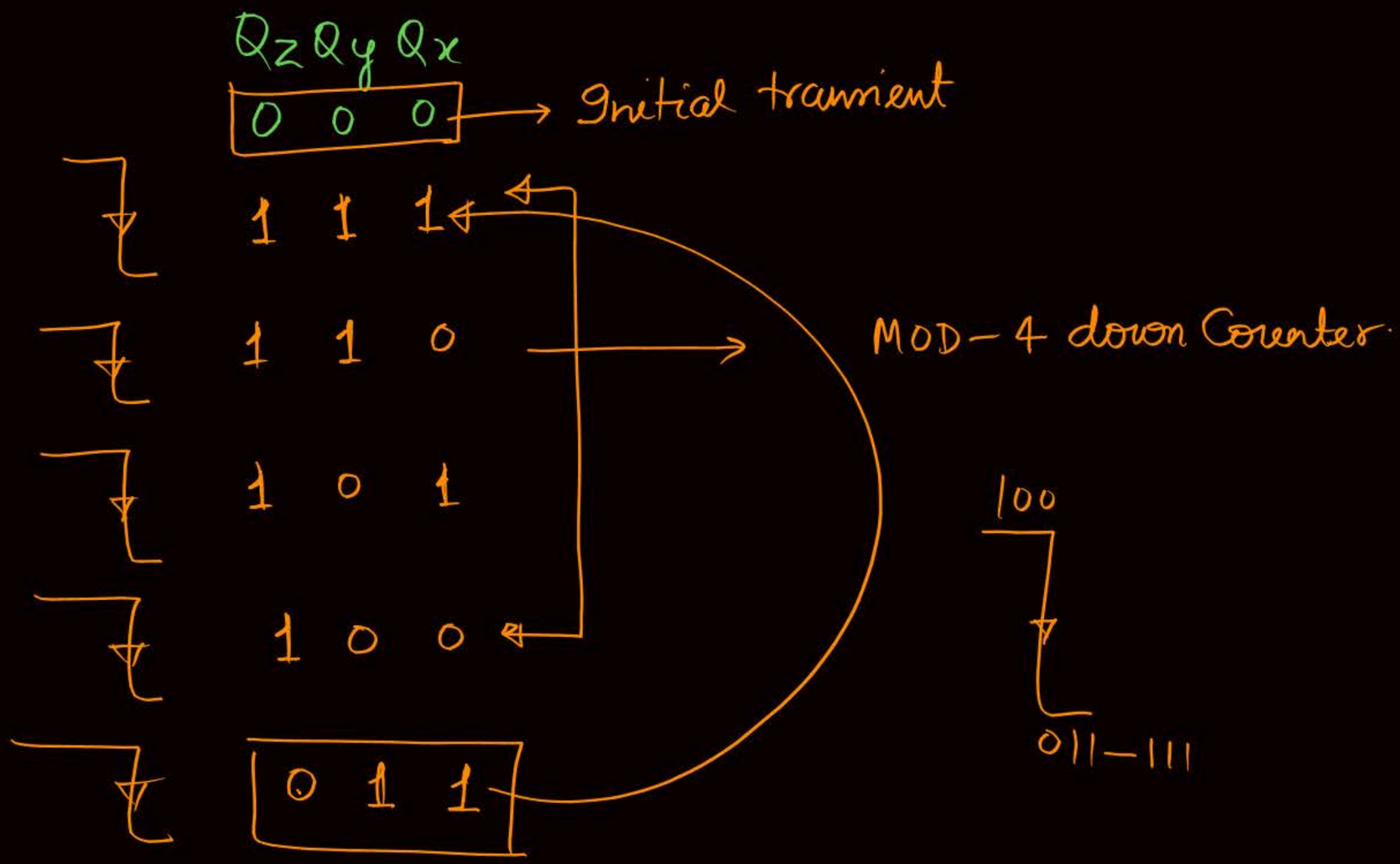
The circuit works as :

(a) MOD-5 down counter

(b) MOD-4 down counter

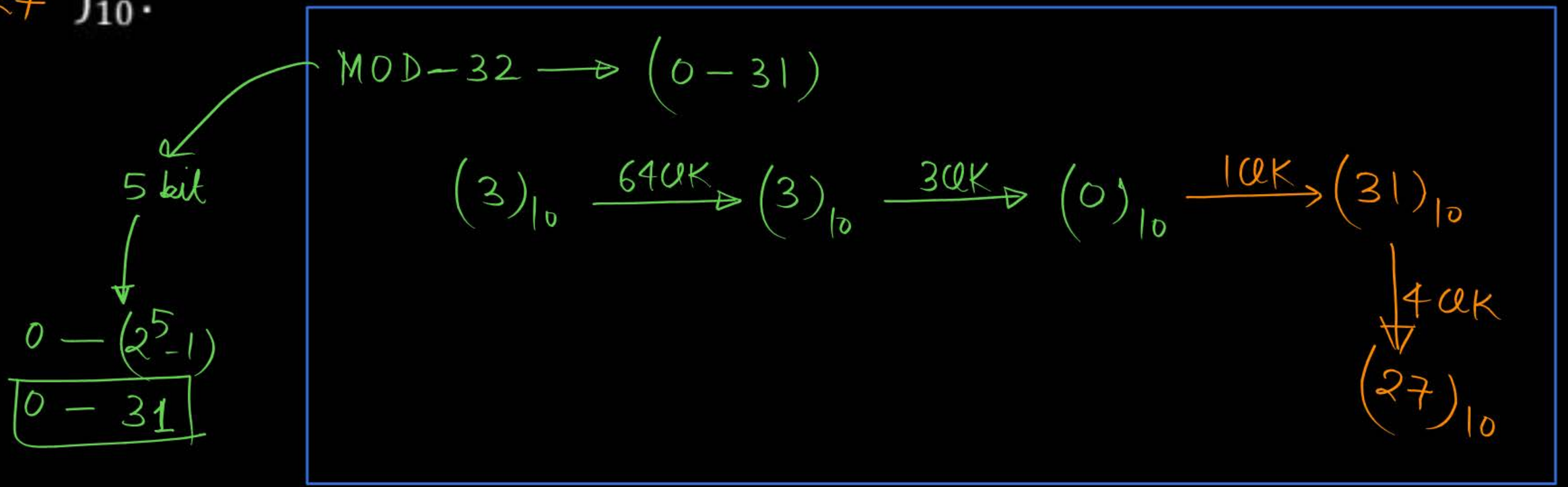
(c) MOD-6 down counter

(d) MOD-5 up counter


QzQyQx
0 0 0
Initial transient
1 1 1
1 1 0
1 0 1
1 0 0
0 1 1
MOD-4 down Counter.
100
011-111

Question

We have a MOD-32 down counter. If its starting state is $(3)_{10}$ then after application of 72 clock pulses the count of the counter will be $(27)_{10}$.



Question

In module-6 ripple counter shown in figure, the output of 2-input gate is used to clear the J-K FF.

Up counter

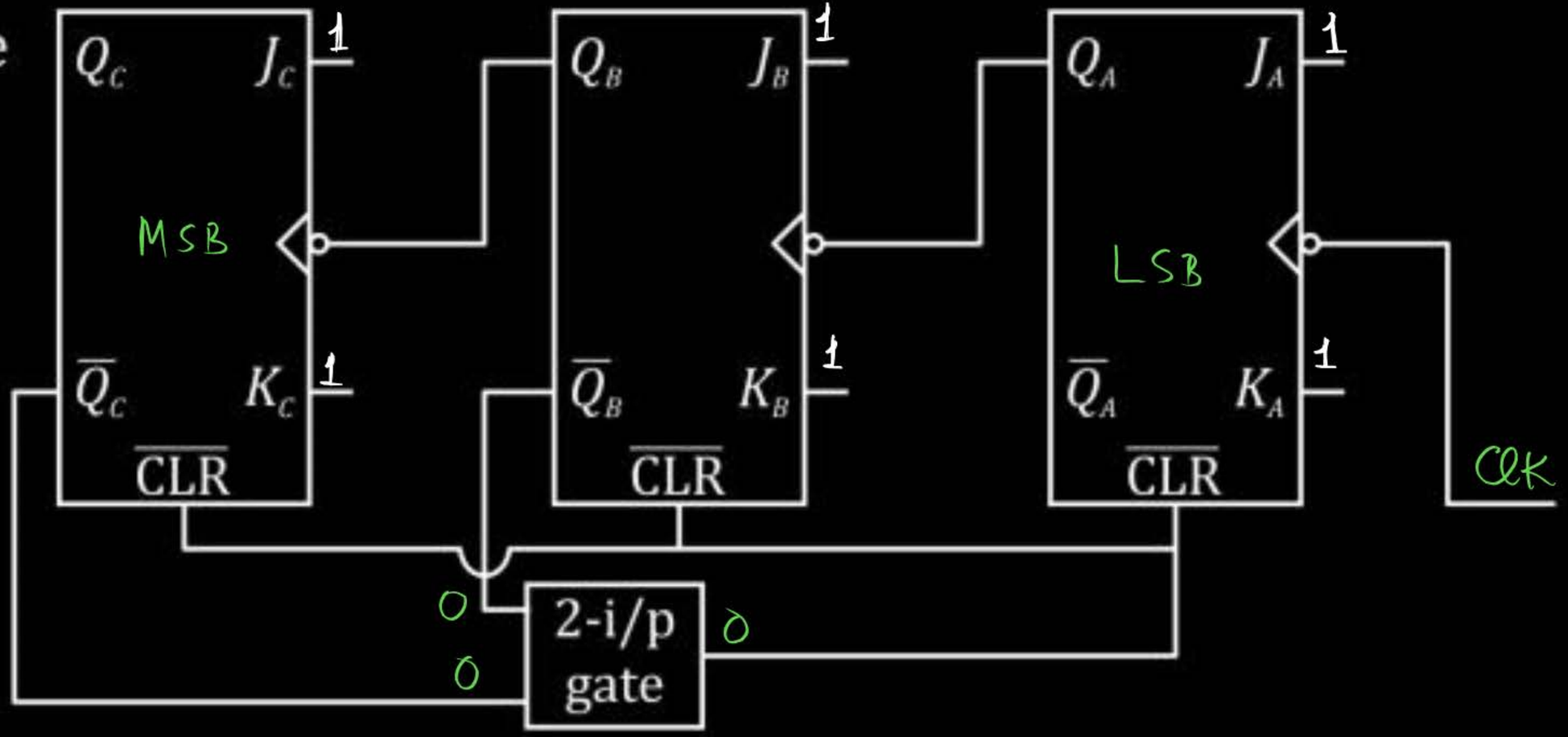
The gate will be

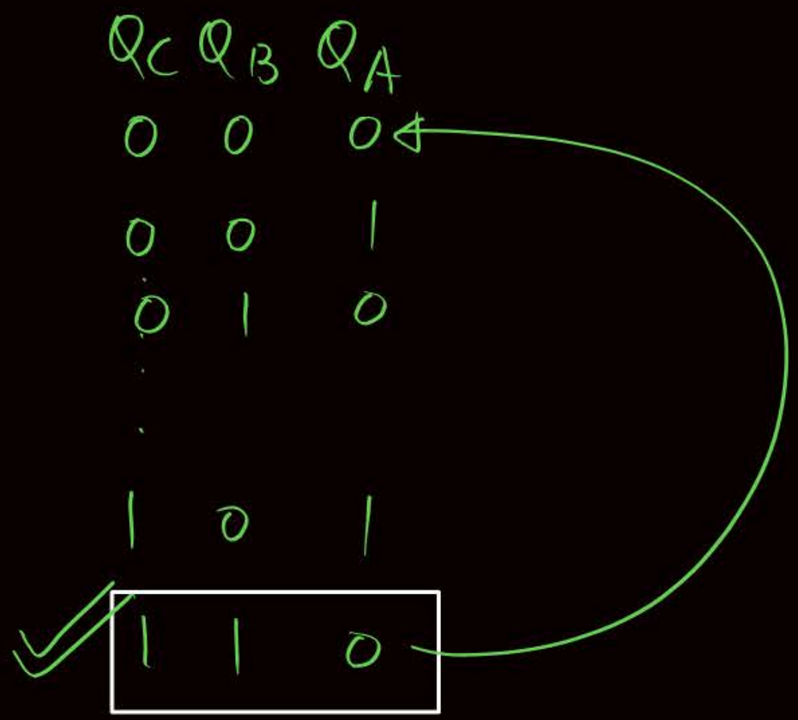
(a) NAND ✗

(b) NOR ✗

(c) OR ✓

(d) ~~AND~~ ✗ ✓




Q_C Q_B Q_A
0 0 0
0 0 1
0 1 0
1 0 1
1 1 0

Question

The figure shows a binary counter with synchronous clear input with decoding logic shown. This counter works as

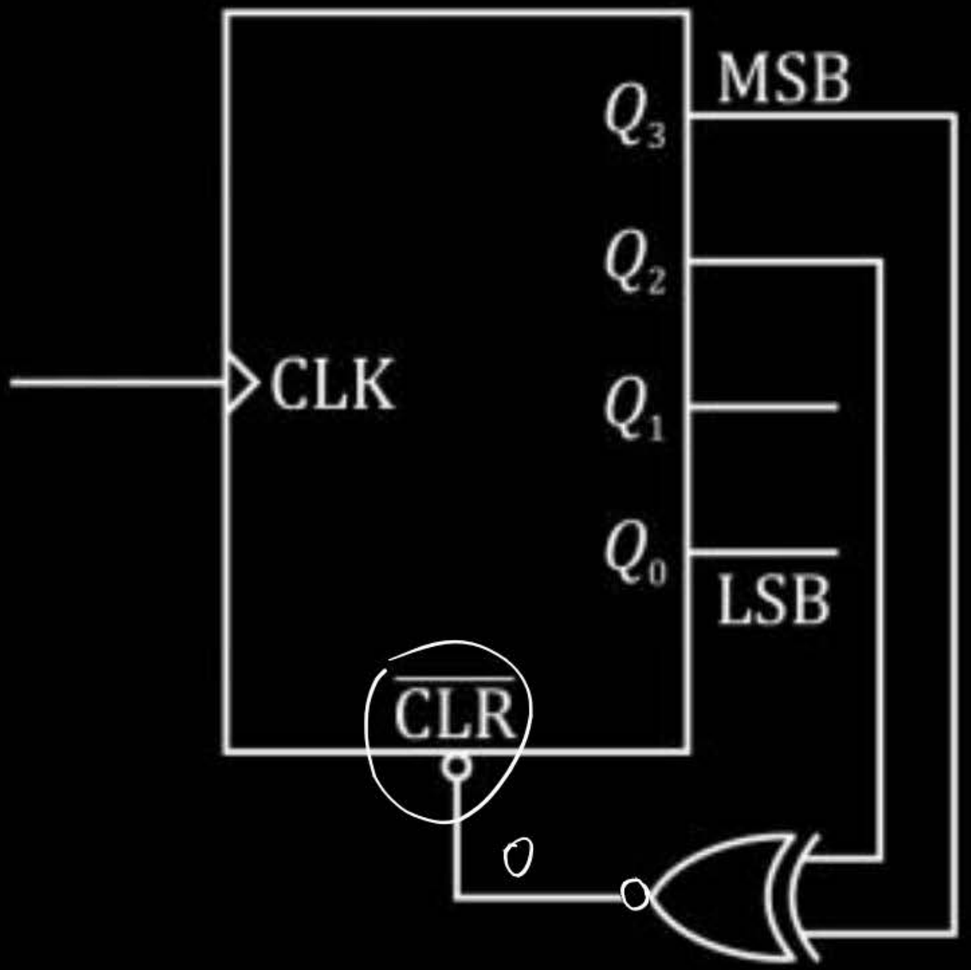
The counter is

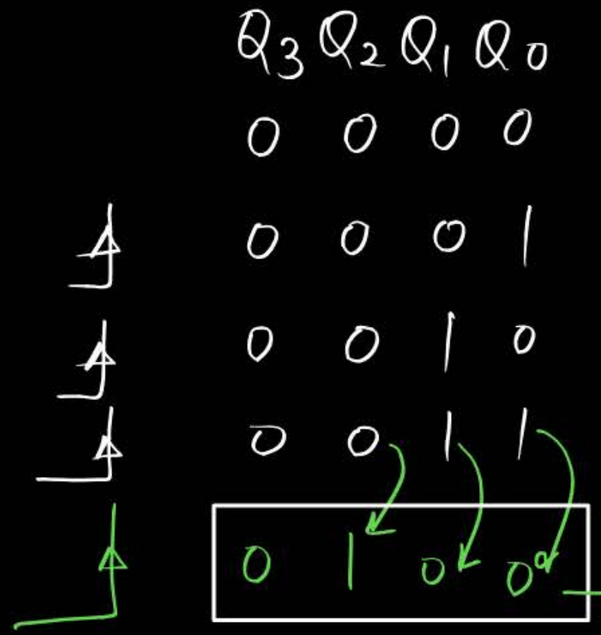
(a) MOD-2 counter

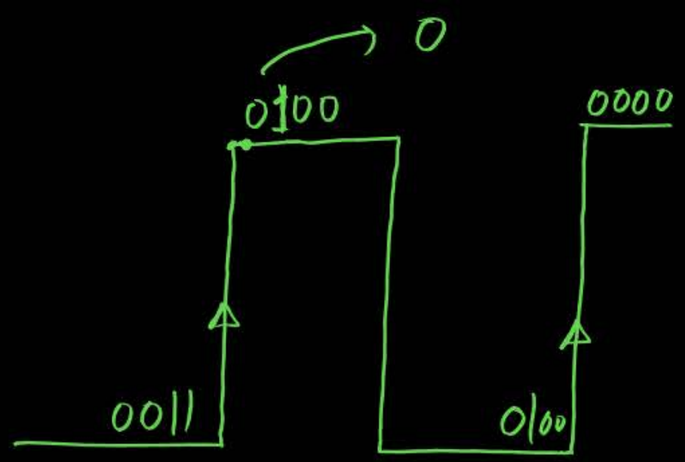
(b) MOD-4 counter

(c) MOD-5 counter

(d) MOD-6 counter







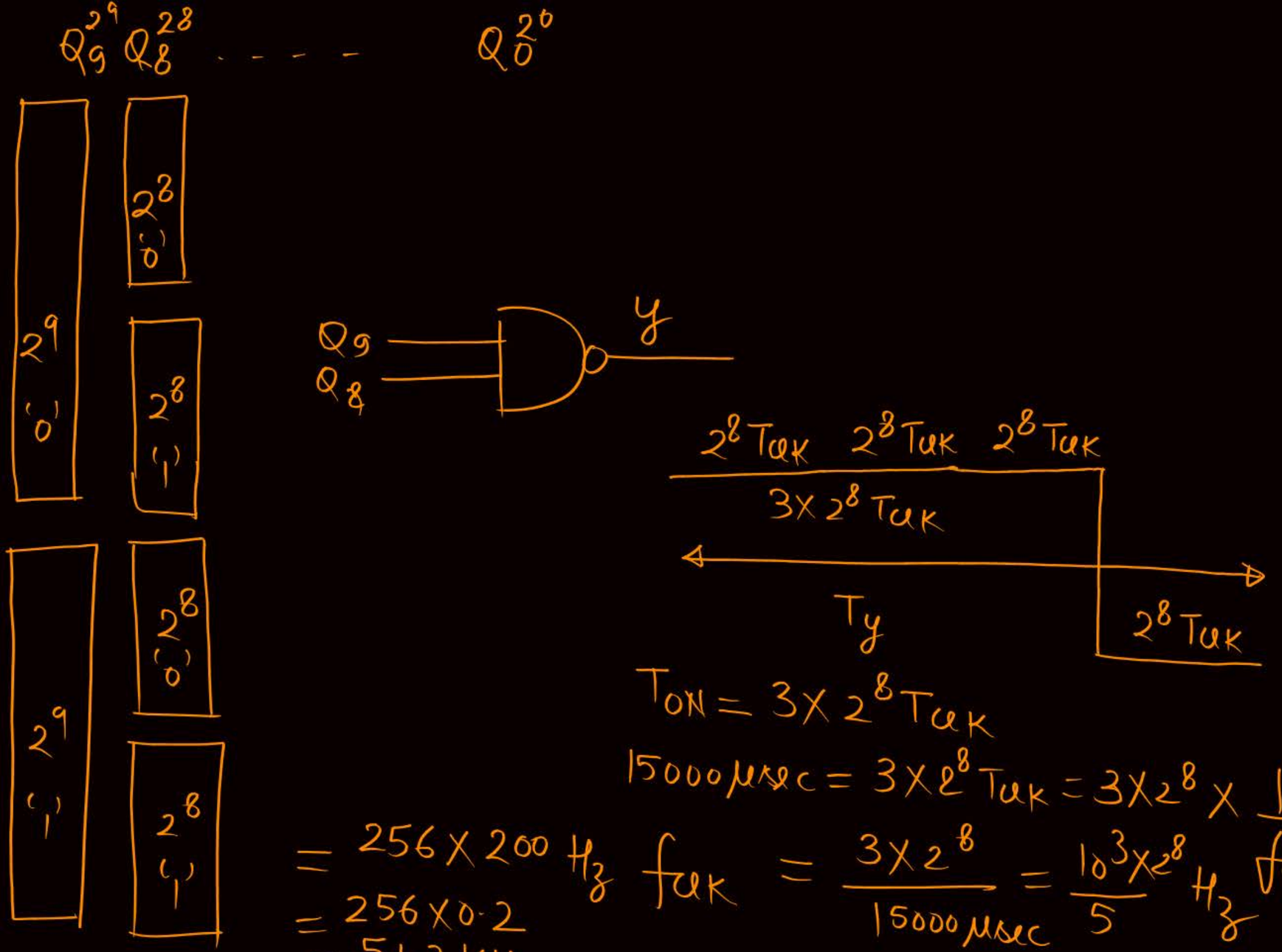
→ Here exist for 1-Tclk so will be counted in MOD no.

Question

We have 10-bit ripple counter which is counting in up sequence. Two most significant bits are NAND together to generate an output y. Output y is periodic waveform with T_{ON} (time duration for which y is high in one period) 15000 μsec. Then the clock frequency (kHz) 512.

- $2^{10} \longrightarrow$ 1024-states $\longrightarrow (0-1023)$ $\quad Q_9 Q_8 \cdots Q_0$





$T_{ON} = 3 \times 2^8 T_{clk}$

$15000\,\mu sec = 3 \times 2^8 T_{clk} = 3 \times 2^8 \times \frac{1}{f_{clk}}$

$f_{clk} = \frac{3 \times 2^8}{15000\,\mu sec} = \frac{10^3 \times 2^8}{5}\,Hz$

$= 256 \times 200\,Hz$

$= 256 \times 0.2$

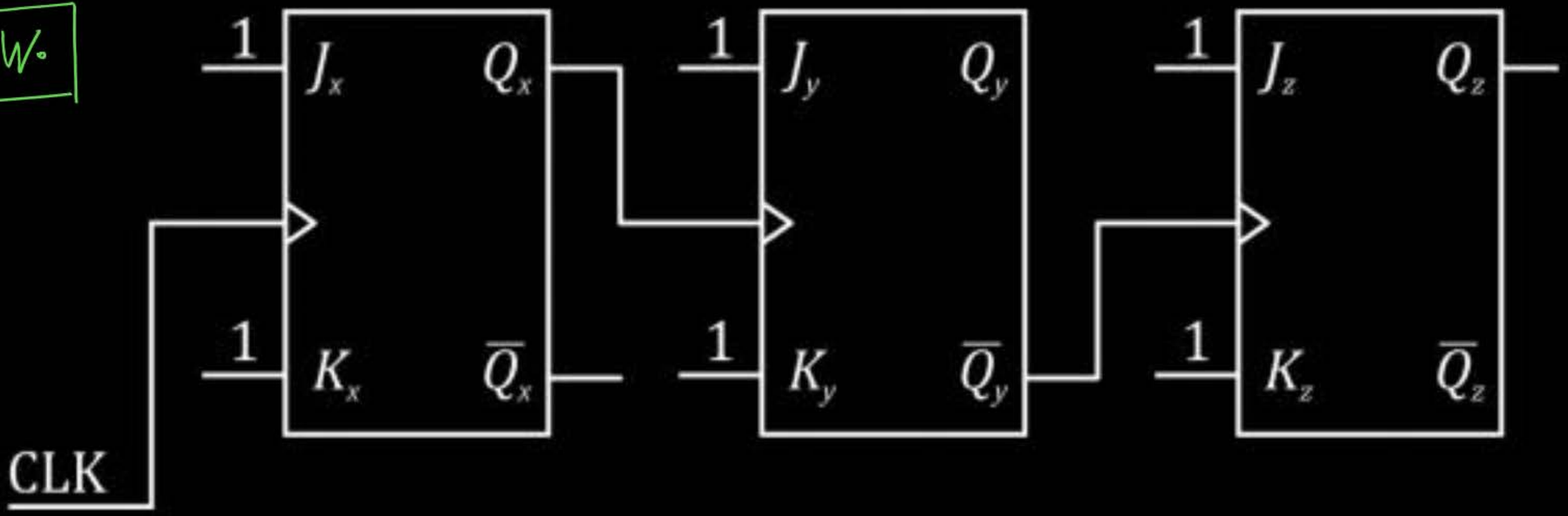
$= 51.2\,KHz$

Question

A sequential circuit is as given below :

MOD-8

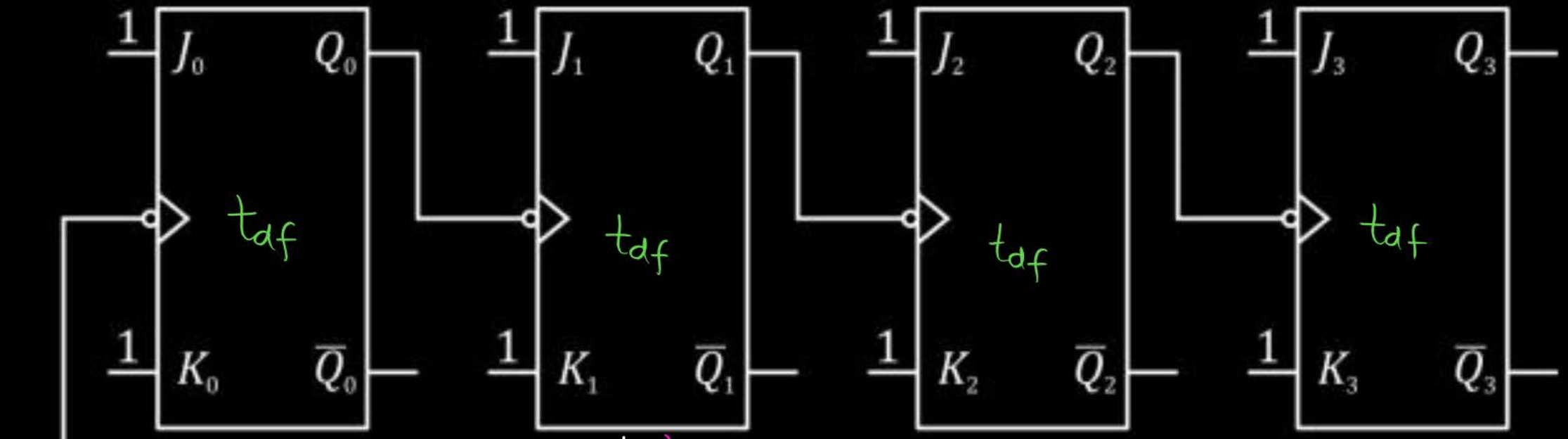
H.W.

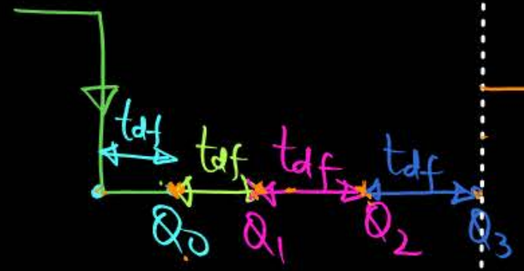


If starting state of the counter $Q_zQ_yQ_x$ is $(101)_2$ then after 251 CLK pulses output $Q_zQ_yQ_x$ is ( )$_{10}$.

Max. Clock Frequency

Lets take a 4-bit asynchronous counter :





→ next trigger.

$T_{CLK} \geq 4\,t_{df}$



ripple Counter.

- IMP Points :

Worst case $\longrightarrow$ $T_d = n t_{df}$

for proper operation of counter

$T_{clk} \geq n t_{df}$

$(T_{clk})_{min} = n t_{df}$

$\frac{1}{T_{clk}} \leq \frac{1}{n t_{df}}$

$\frac{1}{f_{clk}} \geq n t_{df}$

$f_{clk} \leq \frac{1}{n t_{df}}$

$f_{max} = \frac{1}{n t_{df}}$



Topic : 2 Min Summary

Asynch. Counter.

Thank you

GW

Soldiers !

